

Question Paper

Exam Date & Time: 29-Dec-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FIRST SEMESTER B.TECH. DEGREE EXAMINATIONS - DEC 2025 / JAN 2026
SUBJECT: MME 1003/MME_1003 FUNDAMENTALS OF MECHANICAL ENGINEERING

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) 250kg of feed water enters the boiler at a temperature of 27°C . steam at a pressure of 1.5 N/mm^2 (4)
leaves the boiler as 17.5% wet and enters the super heater where it is superheated to a temperature of 310°C . Further, from the super heater if the steam is led to a heat exchanger for process heating where it loses 65.8% of enthalpy, determine the following;
i) Total heat supplied to water in the boiler
ii) Total heat supplied steam in the superheater.
iii) The condition and related parameter of the steam at the exit point of heat exchanger.
Assume the specific heat of water as 4.187 kJ/KgK and that of super-heated steam as 2.25 kJ/KgK .

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 1B) Boilers A and B supply steam to a common pipeline at a ratio of 7:3 at a pressure of 185 N/cm^2 . The (3)
steam from boiler-A is at temperature 280°C , while the condition of steam from boiler-B is unknown.
The steam in the common pipeline is 18% wet at 1.85 MPa . The calorific value of fuel supplied to boiler-A is 39.5 MJ/kg and boiler-B is 37.75 MJ/kg . Compute the condition of steam supplied by boiler-B.
Assume specific heat of superheated steam and saturated water as 2.25 kJ/kgK and 4.187 kJ/kgK respectively.

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 1C) Explain any three boiler accessories, that enhance the efficiency of a boiler. (3)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 2A) A belt drive is used to transmit the power from a DC motor to the head-stock in a lathe machine. DC (4)
motor shaft rotates at a constant speed of 1200 rpm . The centre distance between the shaft of the belt drive is 1.85 m . The speed reduction ratio is 3.5, and the angle of contact on the smaller pulley is 4.15 radians . The larger pulley has a diameter of 80 cm . The initial belt tension is 850 N and the coefficient of friction is 0.25, sketch the arrangement and calculate the following:
i) Power transmitted in the belt drive
ii) Length of the belt drive
iii) Width of the belt drive, if permissible tension in the belt drive is 8 kN per meter width of the belt.
- 2B) The following gear wheels are available to form a compound gear train. Sketch the arrangement of (3)
the gear train using four shafts, to obtain highest velocity ratio at the output shaft and calculate the same. Also mention the conditions used to achieve the condition.

Gear	Type	No. of teeth	Module (mm)
A	Spur	92	2
B	Spur	40	3
C	Spur	30	2
D	Spur	75	2
E	Spur	60	3
F	Spur	45	3
G	Helical	95	3
H	Helical	120	3
I	Helical	35	2
J	Helical	70	2
K	Helical	80	2
L	Helical	30	3
M	Helical	57	2

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 2C) Comment on the following: (3)
- Life of the belt in cross belt drive is lower as compared with open belt drive.
 - Friction is an helpful agent in case of transmission of power using belts
 - Contact area of the belt on the pulley influences the power transmission.

Course Outcome: "CO2" / Cognitive Level: "Application"

- 3A) Illustrating thermodynamic events explain working of an IC engine that directly admits the charge and operates with a compression ratio of 18:1. (4)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 3B) The following observations were recorded during a test on a 4-stroke engine. Bore=15cm, Stroke =20cm, Cam Shaft Speed= 250 rpm, Mean effective pressure= 8bar, Fuel consumption= 0.0013kg/sec, Calorific value of the fuel= 43900kJ/kg. Determine the indicated thermal efficiency. (3)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 3C) Give suitable reasons for the following: (3)
- For the same capacity four stroke engines are heavy and bulkier as compared to two stroke engines.
 - Two stroke petrol engines generate more pollution as compared to four stroke petrol engines.
 - Multi cylinder engine requires a lighter flywheel as compared to a single cylinder engine.

Course Outcome: "CO3" / Cognitive Level: "Application"

- 4A) A simple refrigerant cycle uses R134a. The evaporator and condenser pressures are 0.234 MPa and 1.01MPa respectively. After compression the refrigerant is superheated by 10 °C Determine: (4)
- COP of the cycle
 - Refrigerating capacity if the mass flow is 7 kg/min.
- Assume that at the exit of the evaporator the refrigerant is dry.

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 4B) Write a short note on: (3)

- i) Unit of refrigeration
- ii) Coefficient of Performance

Course Outcome: "CO3" / Cognitive Level: "Understand"

- 4C) Write the functions of the following lathe parts: (3)
- i. Chuck
 - ii. Tailstock
 - iii. Bed

Course Outcome: "CO3" / Cognitive Level: "Understand"

- 5A) Explain Clausius statement and Kelvin Plank statement. (4)

Course Outcome: "CO3" / Cognitive Level: "Understand"

- 5B) List and explain any 3 advantages of CNC machines. (3)

Course Outcome: "CO3" / Cognitive Level: "Understand"

- 5C) What are the various stages in producing a part using additive manufacturing technique. (3)

Course Outcome: "CO3" / Cognitive Level: "Understand"

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Event Name	Subject
058-MIT BLR-MAKEUP EXAM-2025-2026	MME_1003:FUNDAMENTALS OF MECHANICAL ENGINEERING

#	Course Outcome	Description	Marks
1	1	CO 1	17
2	2	CO 2	6
3	3	CO 3	23
Total			46

* **NA** - CO Not Selected

Event Name	Subject
058-MIT BLR-MAKEUP EXAM-2025-2026	MME_1003:FUNDAMENTALS OF MECHANICAL ENGINEERING

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	5	16(34.78%)
3	Application	3	9(19.57%)
4	Analysis	6	21(45.65%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		14	46 (100%)